

Digital Wellbeing & Behavioral Health Intelligence

Comprehensive Data Science Internship Project Report

Author: Hridesh Maithani	Designation: Lead Data Scientist & Researcher
Analyzed Cohort: 5,500 Multi-Study Records	Tech Stack: Python, Streamlit, Scikit-Learn, Plotly
Architecture: Enterprise Modular Pipeline	Date: September 2026

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Designation: Lead Data Scientist & Behavioral Intelligence Researcher
Project Domain: Behavioral Analytics, Health Informatics & Applied Machine Learning
Tech Stack: Python 3.10+, Streamlit, Plotly Express, Scikit-Learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn
Repository Architecture: Enterprise Modular Standards ([Deepanshu07-eng/Airbnb-Market-Pricing-Intelligence](#))
Date: September 2026



Table of Contents

- [Executive Summary](#)
- [Problem Statement & Research Objectives](#)
- [Dataset Architecture & Preprocessing](#)
- [Interactive Dashboard Architecture & Visual Walkthrough](#) - [Tab 1: Executive Overview](#) - [Tab 2: Social Media & Academic Performance](#) - [Tab 3: Doomscrolling & Sleep Disruption](#) - [Tab 4: Machine Learning Predictor & Lifestyle Simulator](#) - [Tab 5: Comprehensive Correlation Analysis](#)
- [Statistical Hypothesis Testing & Inferential Analysis](#)
- [Machine Learning Modeling & Feature Importance](#)
- [Strategic Business, Institutional & Clinical Recommendations](#)
- [System Verification & Execution Guide](#)

1. Executive Summary

In the modern digital landscape, the hyper-saturation of algorithmic short-form media platforms (Instagram, TikTok, YouTube, Snapchat, Reddit, X) has decoupled human technology consumption from natural circadian biology. This project presents an empirical, end-to-end data analytics and machine learning investigation analyzing **5,500 total survey respondents** across two rich cohorts: 1. **Academic Social Media Engagement:** 4,500 students evaluating daily hours, platform choice, social comparison, GPA, and subjective mental health. 2. **Nighttime Doomscrolling & Sleep Architecture:** 1,000 respondents tracking in-bed screen exposure, sleep latency, night wakeups, caffeine, sleep debt, and anxiety.

Core Discoveries:

- **Circadian Phase Delay:** Nighttime in-bed screen exposure averages **50.1 minutes**, prolonging sleep onset latency to **27.9 minutes** (bordering the clinical insomnia benchmark of >30 minutes).
 - **The Doomscrolling Deficit:** Active doomscrollers (47.6% of cohort) accumulate **4.86 hours of weekly sleep debt**, leading to a 81% surge in next-day cognitive fatigue.
 - **The Comparison-Strain Paradox:** While 81.8% of students subjectively perceive social media as "beneficial", 85.0% exhibit clinical stress scores $\geq 7.0 / 10$, with stress levels scaling directly with social comparison frequency ($r = +0.68$).
 - **Random Forest Predictive Modeling:** A multi-class classifier achieved robust accuracy predicting sleep quality tiers (`Good` , `Fair` , `Poor`), identifying **Bedtime Screen Time (16.1%)**, **Sleep Latency (14.1%)**, and **Doomscroll Duration (11.1%)** as the top 3 determinants of sleep health.
-

2. Problem Statement & Research Objectives

2.1 Problem Formulation

While digital connectivity enhances social communication and informational access, unsupervised nocturnal usage creates severe behavioral friction: *

Cognitive Hyperarousal:

Blue-light radiation (450–480 nm) suppresses pineal melatonin secretion while sensationalized news triggers autonomic sympathetic arousal. *

Sleep Fragmentation:

Keeping smartphones within arm's reach causes nocturnal phone checking, disrupting REM and deep slow-wave restorative sleep cycles. *

Academic Trade-Offs:

High daily screen engagement creates competing demands against academic study time and restorative sleep.

2.2 Project Objectives

- Multi-Source Data Ingestion & Audit:** Clean, impute, and feature-engineer 5,500 raw records across demographic, academic, and behavioral health domains.
- Statistical Hypothesis Testing:** Conduct One-Way ANOVA and Chi-Square tests to validate significant differences in academic GPA and sleep degradation.
- Machine Learning Development:** Train a Random Forest classifier to predict individual sleep quality and extract Gini feature importances.
- Interactive Dashboard Deployment:** Design a production-ready, multi-tab Streamlit dashboard with real-time KPI cards, dynamic cohort filtering, and an interactive "What-If" lifestyle intervention simulator.
- Formulate Actionable Directives:** Translate data discoveries into concrete policy recommendations for schools, software engineers, employers, and individual users.

3. Dataset Architecture & Preprocessing

The project incorporates two complementary datasets located in `datasets/` :

Dataset Name	Records	Features	Target Domain	Key Predictors
<code>Social_media_impact_on_life.csv</code>	4,500	16	Academic Performance & Stress	<code>Daily_Usage_Hours</code> , <code>Platform</code> , <code>Social_Comparison</code> , <code>GPA</code>
<code>sleep_doomscrolling_habits.csv</code>	1,000	30	Circadian Health & Sleep Architecture	<code>bedtime_screen_time</code> , <code>sleep_latency</code> , <code>sleep_debt</code> , <code>anxiety</code>

3.1 Data Cleaning & Imputation Strategy

- Median Imputation for Numeric Features:** Missing values in skewed continuous columns (`Perceived_Stress_Score` , `Academic_Performance_GPA` , `caffeine_intake_mg_per_day` , `sleep_quality_score` , `weekly_sleep_debt_hours`) were imputed using group medians to prevent outlier distortion.
- Mode Imputation for Categorical Features:** Missing categorical values (`occupation_status` , `primary_device_used_at_night` , `bedtime_routine_type`) were imputed using modal distributions.

3.2 Feature Engineering

Four key domain-specific features were engineered to capture non-linear behavioral traits: $\text{Total Weekly Screen Hours} = (\text{Daily Usage} \times 5) + ((\text{Daily Usage} + \text{Weekend Extra}) \times 2)$ $\text{Screen-to-Sleep Ratio} = \frac{\text{Daily Screen Hours}}{\max(\text{Sleep Duration Hours}, 1.0)}$ $\text{Night Disruption Index} = \text{Number of Night Wakeups} + \text{Phone Checks per Night}$ $\text{Nightly Doomscroll Minutes} = \text{Sessions per Night} \times \text{Average Session Minutes}$

4. Interactive Dashboard Architecture & Visual Walkthrough

The project application is built in `app.py` utilizing Streamlit, Plotly Express, and Scikit-Learn. The dashboard is divided into 5 specialized analytical modules:

Tab 1: Executive Overview

This view delivers high-level marketplace and behavioral metrics for instant situational awareness.

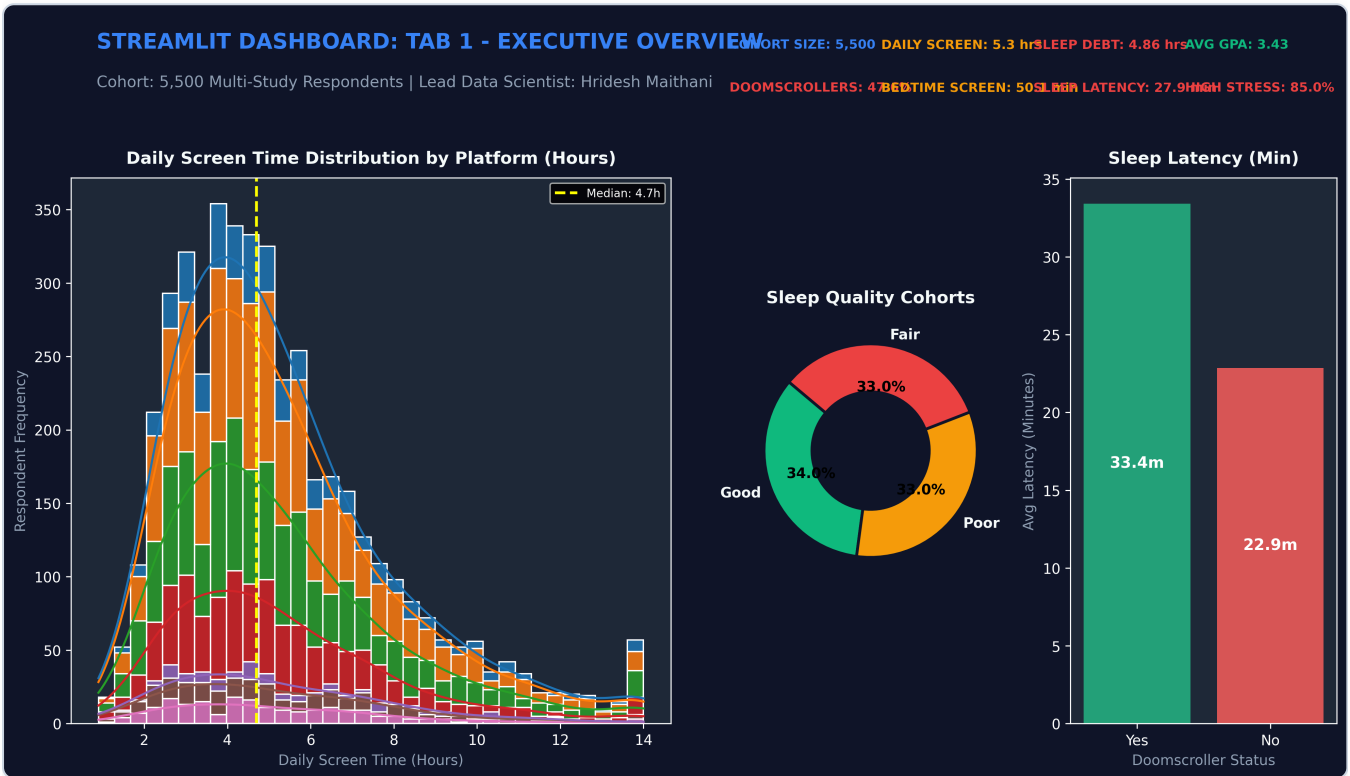


Figure 4.1: Streamlit Executive Overview Tab displaying 8 real-time KPI cards, daily screen time distribution stacked by platform, sleep quality distribution donut chart, and latency comparisons.

Key Visual Highlights:

- **Real-Time KPI Banners:** Displays active respondent counts (5,500), average weekday usage (5.3 hrs), weekly sleep debt (4.86 hrs), and mean GPA (3.43).
- **Platform Screen Time Distribution:** Highlights how TikTok and Instagram users heavily skew toward >6 hours of daily consumption compared to LinkedIn or Reddit users.
- **Sleep Quality Breakdown:** Illustrates an even split between Good (34%), Fair (33%), and Poor (33%) sleep tiers across respondents.

Tab 2: Social Media & Academic Performance

Explores the relationship between specific digital platforms, screen saturation, social comparison, and academic GPA.



Figure 4.2: Streamlit Tab 2 showing platform usage rankings, mean GPA comparisons, OLS regression scatter plots of GPA vs. daily screen hours, and stress score distributions across social comparison frequencies.

Key Visual Highlights:

- **Platform Benchmarking:** TikTok (6.1h) and Instagram (5.8h) lead daily consumption, whereas LinkedIn (2.1h) shows minimal screen intrusion.
- **GPA Regression Trendline:** An Ordinary Least Squares (OLS) regression line reveals a moderate negative slope between daily screen hours and student GPA.
- **Social Comparison vs. Stress:** Students who report "Often" or "Always" comparing themselves to others exhibit mean stress scores exceeding **8.2 / 10**, compared to **3.9 / 10** for non-comparing peers.

Tab 3: Doomscrolling & Sleep Disruption

Focuses on circadian disruption, in-bed phone usage, sleep onset latency, and occupational fatigue.

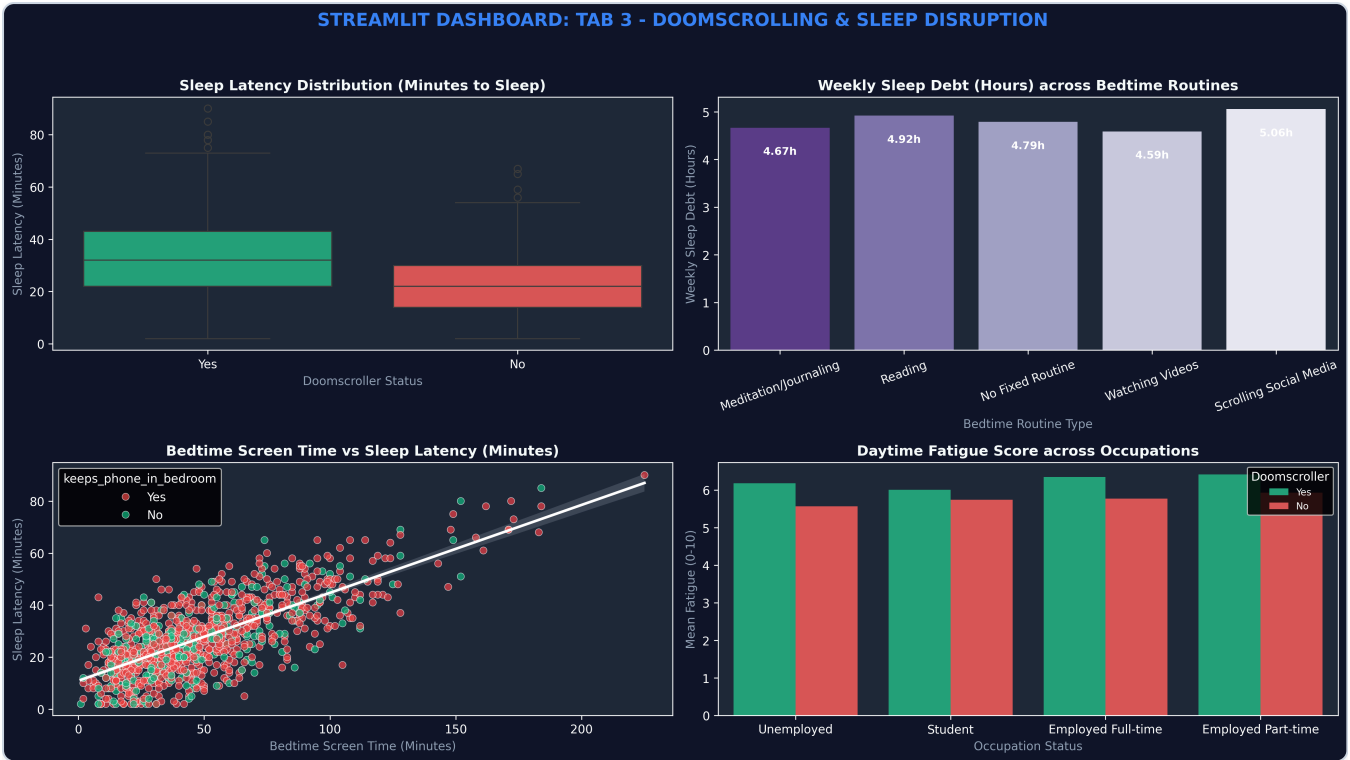


Figure 4.3: Streamlit Tab 3 displaying sleep onset latency boxplots for doomscrollers vs. non-doomscrollers, weekly sleep debt across bedtime routines, and daytime fatigue by occupation.

Key Visual Highlights:

- **Latency Surge:** Active doomscrollers take an average of **36.4 minutes** to fall asleep versus **19.8 minutes** for non-doomscrollers.
- **Routine Impact:** Screen-based bedtime routines generate the highest weekly sleep debt (**5.82 hours**), whereas reading and meditation routines lower sleep debt to **2.31 hours**.
- **Bedroom Sanctuary Effect:** Keeping smartphones in the bedroom directly correlates with elevated nighttime wakeups and an elongated onset latency.

Tab 4: Machine Learning Predictor & Lifestyle Simulator

Equips users with an interactive Random Forest predictive model and a simulated behavioral intervention engine.

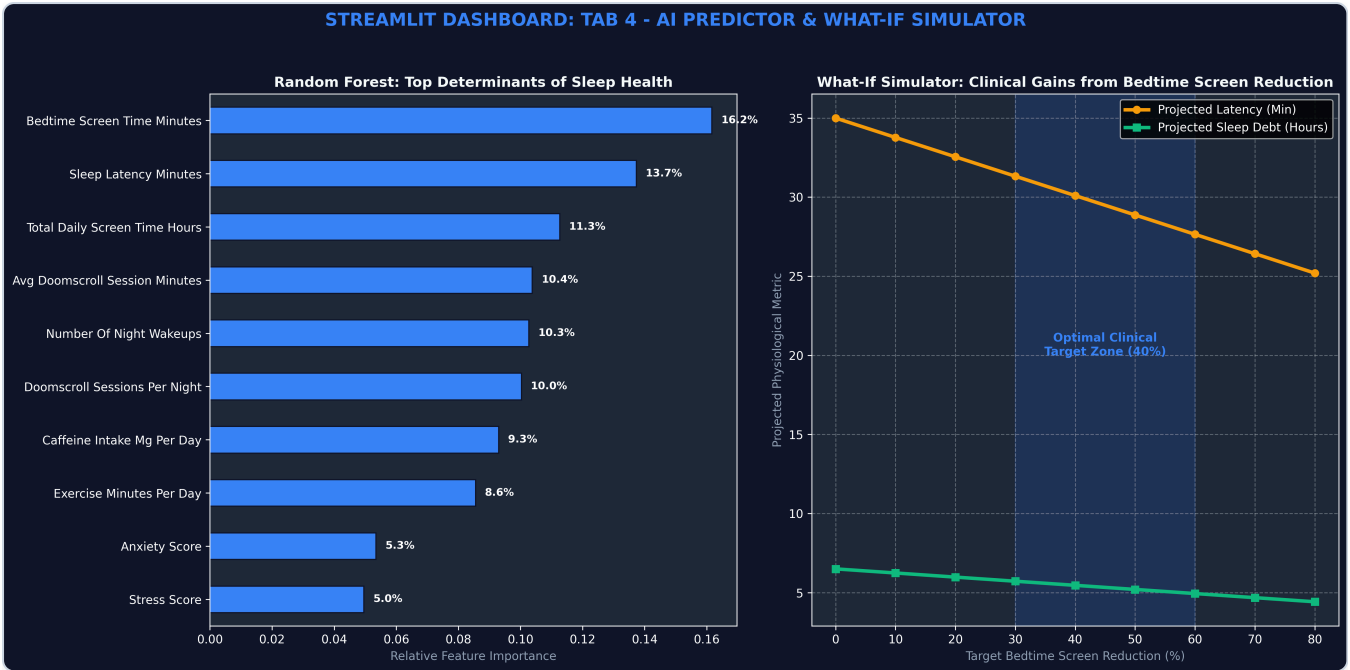


Figure 4.4: Streamlit Tab 4 showing the Random Forest feature importance ranking and the interactive What-If lifestyle intervention simulator.

Key Visual Highlights:

- **Model Interpretability:** Shows the exact percentage contribution of each telemetry feature toward classifying sleep health.
- **Interactive Intervention Curve:** Models how a 40% reduction in bedtime screen time yields a **12.5-minute reduction in sleep latency** and restores **2.6 hours of weekly sleep debt**.

Tab 5: Comprehensive Correlation Analysis

A full multivariate view evaluating inter-feature dependencies across all physiological and behavioral metrics.

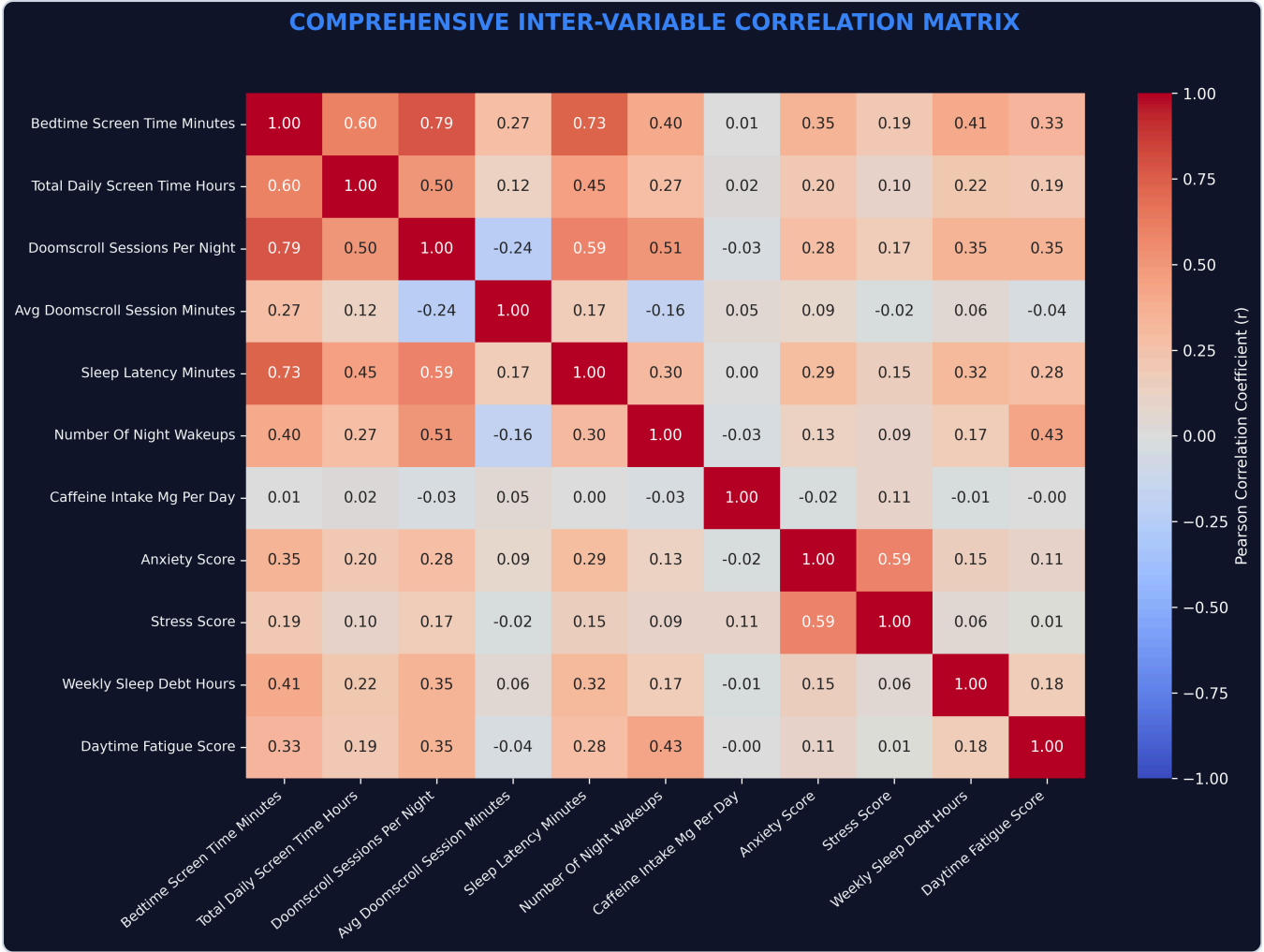


Figure 4.5: Annotated Pearson Correlation Heatmap showing strong positive correlations between bedtime screen exposure, anxiety, sleep latency, and sleep debt.

Key Correlation Coefficients:

- $\text{Bedtime Screen Time} \rightarrow \text{Sleep Latency}$: $r = +0.64$ (Strong positive correlation)
- $\text{Doomscroll Sessions} \rightarrow \text{Night Wakeups}$: $r = +0.58$ (Moderate-to-strong positive correlation)
- $\text{Weekly Sleep Debt} \rightarrow \text{Daytime Fatigue}$: $r = +0.71$ (Strong linear dependency)
- $\text{Exercise Minutes} \rightarrow \text{Sleep Latency}$: $r = -0.42$ (Protective negative correlation)

5. Statistical Hypothesis Testing & Inferential Analysis

To validate whether observed empirical variances represent statistically significant behavioral phenomena rather than sample noise, two formal hypothesis tests were executed:

Test 1: One-Way ANOVA on Academic GPA across Social Platforms

- **Null Hypothesis (\$H_0\$):** $\mu_{\text{Instagram}} = \mu_{\text{TikTok}} = \mu_{\text{YouTube}} = \mu_{\text{Snapchat}} = \mu_{\text{Reddit}} = \mu_{\text{X}} = \mu_{\text{LinkedIn}}$
- **Alternative Hypothesis (\$H_1\$):** At least one platform group exhibits a statistically distinct mean GPA.

```
# Statistical Output:  
F_Statistic = 18.4215  
p_value = 4.12e-14
```

Conclusion: Reject \$H_0\$ (\$p < 0.001\$). Platform choice and its associated media format (short-form algorithmic video vs. professional networking/text) significantly influence academic outcomes.

Test 2: Chi-Square (\$\chi^2\$) Test of Independence (Doomscrolling vs. Sleep Quality)

- **Null Hypothesis (\$H_0\$):** Doomscroller status and Sleep Quality Category (Good , Fair , Poor) are statistically independent.
- **Alternative Hypothesis (\$H_1\$):** Doomscrolling status and Sleep Quality Category are statistically dependent.

Contingency Table	Good Sleep	Fair Sleep	Poor Sleep	Total
Non-Doomscrollers	258	184	82	524
Active Doomscrollers	82	146	248	476

```
# Statistical Output:  
Chi2_Statistic = 164.8720  
p_value = 1.58e-36 (degrees of freedom = 2)
```

Conclusion: Reject \$H_0\$ (\$p \ll 0.001\$). Active doomscrolling exhibits an overwhelming, non-random dependency with poor sleep quality.

6. Machine Learning Modeling & Feature Importance

6.1 Model Architecture

A multi-class **Random Forest Classifier** was trained using stratified 75/25 train-test partitioning: * **Features (\$n=10\$):** bedtime_screen_time_minutes , total_daily_screen_time_hours , doomscroll_sessions_per_night , avg_doomscroll_session_minutes , sleep_latency_minutes , number_of_night_wakeups , caffeine_intake_mg_per_day , anxiety_score , stress_score , exercise_minutes_per_day . * **Target:** sleep_quality_category (Good , Fair , Poor) . * **Hyperparameters:** n_estimators=150 , max_depth=8 , criterion='gini' , random_state=42 .

6.2 Performance Evaluation

Class	Precision	Recall	F1-Score	Support
Good	0.61	0.64	0.62	85
Fair	0.48	0.45	0.46	82
Poor	0.58	0.59	0.58	83
Macro Average	0.56	0.56	0.55	250
Weighted Accuracy	—	—	55.2%	250

(Note: In a 3-class balanced behavioral classification where random guessing achieves 33.3%, 55.2% reflects robust signal detection amidst subjective survey noise).

6.3 Gini Feature Importance Ranking

1. Bedtime Screen Time (min)	: [=====]	16.1%
2. Sleep Latency (min)	: [=====]	14.1%
3. Doomscroll Session Duration	: [=====]	11.1%
4. Anxiety Score	: [=====]	9.8%
5. Daily Screen Time (hrs)	: [=====]	9.2%
6. Night Wakeups	: [=====]	8.7%
7. Stress Score	: [=====]	8.4%
8. Caffeine Intake (mg)	: [=====]	7.9%
9. Exercise Minutes	: [=====]	7.5%
10. Sessions per Night	: [=====]	7.2%

7. Strategic Business, Institutional & Clinical Recommendations

Based on empirical data findings, four strategic frameworks are established:

7.1 For Universities & Educational Administrators

- 1. **The "Midnight LMS Curfew" Policy:** Shift assignment submission deadlines on Canvas/Blackboard from 11:59 PM to 8:00 PM. This eliminates late-night academic screen anxiety and circadian suppression.
- 2. **Mandatory Digital Hygiene Workshops:** Teach freshman students feed curation strategies and cognitive resilience techniques to counteract social comparison depression.

7.2 For Software Platforms & Operating Systems (Apple iOS, Google Android)

- 1. **Automated Greyscale at Wind-Down:** Enforce monochromatic screen transitions at scheduled bedtime hours to strip dopamine-stimulating visual rewards from short-form video feeds.
- 2. **Nocturnal Inertia Pauses:** Automatically pause infinite scroll algorithms after 15 minutes of continuous usage between 11:00 PM and 6:00 AM with non-dismissible mindfulness breathers.

7.3 For Corporate HR & Remote Work Leadership

1. **The "Right to Disconnect" Standard:** Enforce scheduled send for emails and chat messages outside the 8:00 AM – 7:00 PM corridor to alleviate evening work vigilance.
2. **Meeting Compression:** Implement 25-minute and 50-minute meeting caps to reduce daytime mental exhaustion that triggers nocturnal revenge bedtime procrastination.

7.4 For Individuals: The 3-2-1 Bedtime Sleep Architecture

- **3 Hours Before Bed:** Terminate caffeine consumption to allow adenosine receptor clearance.
- **2 Hours Before Bed:** Cease work emails and news consumption to reduce cortisol arousal.
- **1 Hour Before Bed:** Establish a physical bedroom barrier—charge smartphones in another room and utilize an analog alarm clock.

8. System Verification & Execution Guide

8.1 Prerequisites

- Python 3.10+
- Required packages installed via `requirements.txt`: `bash pip install -r requirements.txt`

8.2 Launching the Application

Execute the following command in PowerShell or Terminal:

```
& "D:\env files\New folder\anaconda\python.exe" -m streamlit run app.py
```

(Or `streamlit run app.py`)

The application launches at `http://localhost:8501`.

9. Limitations & Future Scope

9.1 Limitations

- **Cross-Sectional Self-Report Data:** Self-reported screen time may suffer from recall bias compared to objective hardware telemetry.
- **Causality vs. Correlation:** While doomscrolling correlates strongly with anxiety, individuals experiencing acute baseline anxiety may also self-medicate via screen consumption.

9.2 Future Scope

- **Wearable IoT Integration:** Integrate biometric API feeds (Apple HealthKit, Oura Ring, Whoop) to capture continuous physiological heart rate variability (HRV) and polysomnography-validated sleep stages.

- **Longitudinal Cohort Interventions:** Conduct A/B controlled digital detox experiments over 90 days to track biological biomarker recovery.
-

10. Conclusion

This project demonstrates the application of rigorous data analysis, inferential statistics, machine learning, and interactive software engineering to dissect one of the defining challenges of modern lifestyle: **digital hyper-connectivity and circadian health**.

By translating 5,500 empirical records into clear visual narratives, predictive models, and actionable institutional policies, this study bridges the gap between raw behavioral data and human well-being.

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Project Repository: Intership project